

Development and Characterization of Textile-Electrode Electroencephalogram Systems for Brain-Computer Interfaces

Conventional electroencephalogram (EEG) systems rely on wet electrodes that require conductive gel, which limits comfort and mobility, however, in recent years, there is growing interest in developing wearable EEG collection devices. Textile-based electrodes have emerged as a promising alternative for wearable EEG systems as they offer both improved comfort, flexibility in use case, and simple integration into fabric for brain-computer interface (BCI) applications [1], [2]. However, a significant challenge posed by these textile-based electrodes is achieving high signal quality comparable to clinical and research grade electrodes due to high impedance across skin and hair connections [1], [3].

The objective of this study is to advance the development of a wearable EEG system using textile-based electrodes for signal acquisition over hair-covered regions and regions where the skin is exposed directly. The project encompasses both hardware and software development, with a focus on prototyping pre-amplifier circuits for textile electrodes and designing computational methods for processing these recorded textile-based EEG signals. An optimized pre-amplifier circuit design and tailored digital signal processing algorithms should greatly improve the signal-to-noise ratio and classification accuracy of textile-based EEG readings, making them comparable to standard clinical and research grade systems.

This project approach follows the iterative design and evaluation of a wearable EEG system using textile-based electrodes with a pre-amplifier circuit that will be prototyped with various hardware filters such as a low-pass filter or unity gain to optimize signals acquired from textile electrodes, following principles established in biopotential amplifier designs which produces low electrical interference [4], [5]. Measurements will be made for input impedance, common-mode rejection ratio, and baseline noise using a signal generator while EEG data will also be digitally filtered through signal processing techniques including adaptive filtering and independent component analysis (ICA) to remove any artifacts [1], [6].

The pre-amplifier circuit will be tested to specifically improve signal acquisition from textile electrodes by maximizing its input impedance and minimizing noise with off-the-shelf components. Afterwards, digital signal processing methods including adaptive filtering and ICA will be implemented to enhance textile-based EEG signals. Models will be developed to mitigate noise, improve signal-to-noise ratio, and optimize classification performance in BCI applications. The combined hardware and software system's performance will be characterized and benchmarked against gold-standard clinical EEG electrodes in both simple experimental paradigms such as eyes-open/eyes-closed and established BCI tasks from both hair-covered and bare-skin regions of the head.

This research aims to demonstrate the technical feasibility of textile-based EEG for wearable BCI applications, and its research outcomes are expected to contribute both to academic knowledge and to potential commercialization efforts in collaboration with industry partners. Findings will support the preparation of a publication detailing textile electrode validation, EEG signal pre-amplifier design, and software-based signal filtering, with future applications extending toward wearable BCI systems and clinical use cases.

References

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